



Conway's Game of trivago

JavaScript Software Engineer, Hotel Search

The Hotel Search Frontend Team is responsible for the JavaScript code base of our main product: **The trivago website.**

We use our knowledge and skills to improve the user experience on our website by implementing exciting new features that our users will love or by improving the performance and flow of our website to ensure our diverse users will receive the best possible user experience.

As part of the Hotel Search team, we work on a daily basis with our Product Owners, Designers, UX Researchers, Business Analysts and others to deliver the best quality product.

We work in flexible project teams that allow everyone to freely choose the most meaningful way to invest their time. Our developers are free to choose the project they want to work on or to work on our Backlog.

This assignment is designed to give you a glimpse of some of the challenges you will be facing in this role. Please be aware there are no perfect solutions - for us, it's more important to see how you find solutions, process your ideas, structure your thoughts and how you make your decision paths.

Be creative but realistic about what's possible. We are thrilled to get to know a bit more about the way you solve tasks.

Good luck!

CHALLENGE

The case study consists of **4 Tasks**, the first 2 are mandatory and the last 2 are optional.

The mandatory tasks build upon each other, so they need to be processed in order to have visual feedback while running the application.

Each task allows you to highlight a different area of your abilities. The optional tasks require advanced knowledge and experience on functional refactoring and we generally don't expect you to be able to solve them.

On your delivery please specify which tasks you've worked on.

If you started with a task but didn't manage to finish it, it might be a good idea to still submit your changes and explain what kept you from finishing it.



Preface

The case study is a game that you can run in your browser once you compiled the source code. It is a variation of the famous 'Conways' Game of Life'.

The trivago version adds a rule-chain to the game that makes it possible to add additional rules easily.

The Tasks

WEB-101: Find and fix the bug in the application

After you have run the application and started it by pressing the 'Power' button on the TV, you will notice, that after two cell generations the screen is empty.

There seems to be a logical bug in the application applying the game's rules so that we can have proper cell evolution visible that lasts for hundreds of generations or more.

Please find and fix the bug.

WEB-102: Finish multi-color support implementation

Make usage of the rule-chain of this game and finish the implementation of an additional rule that we call 'getMostFrequentColor'.

It should be capable of collecting the most frequent color of a cell's neighbors and apply it to the current cell.

Please finish the implementation of the 'getMostFrequentColor' rule.

WEB-103: A nicer reducer (OPTIONAL)

Have a closer look at the applications reducer. It is a typical redux reducer, but for our taste, there are some method calls, that look kind of similar.

Please refactor the reducer in a way that redundant code snippets can be avoided.

WEB-104: Remove further code duplication by refactoring (OPTIONAL)

The view index makes use of two helper function 'actionCreator' and 'dispatchTo'. They are already abstracting a lot of duplication out of the code, but now these methods are being called over and over again with similar arguments.

Please think about a strategy to remove duplication and refactor the code that calls these methods.